

Analysis of LKA Performance

Greatest Deviation Zones

The highest lane deviations were mainly observed while entering curved sections and during transitions from straight roads to curves. In these regions, the lane position changes quickly in the camera view, making it harder for the controller to respond instantly.

Small deviations were also caused by:

- rapid curvature changes,
 - slight lane detection fluctuations,
 - steering response delay during sharp turns.
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Algorithm Response to Curvature

The LKA system was able to handle curved roads using continuous camera-based lane detection and PID steering correction.

As the vehicle entered a curve:

- the lane error increased,
- the controller generated stronger steering corrections,
- the vehicle gradually returned toward the lane center.

The single-lane tracking approach improved stability on curves by reducing issues related to midpoint estimation and perspective distortion.

Straight Road Behavior

On straight roads, the lane error remained smaller and the steering corrections became minimal. The controller continuously made small adjustments to maintain alignment within the lane.

Minor steering oscillations were observed due to:

- camera noise,
- slight variations in lane detection,
- continuous closed-loop feedback correction.

This is normal behaviour in real-time Lane Keeping Assist systems.

Steering Response Analysis

The steering controller responded dynamically to lane deviation:

- positive steering values corrected the vehicle in one direction,
- negative steering values corrected it in the opposite direction.

The steering smoothing mechanism helped reduce sudden steering changes and improved overall vehicle stability.

Overall System Performance

The implemented system successfully demonstrated:

- real-time lane detection,
- camera-based lane tracking,
- PID-based steering correction,
- stable lane following on straight and curved roads.

The entire steering control was based only on camera input, while the road sensor data was used only for evaluation and performance analysis.